

Solution Attribute Analysis — K = 2 and K = 3 (V2)

Supplemental Material for: "A Data Science-Based, Tactical Role Independent, Comprehensive Metric to Evaluate Professional Football Players"

This document presents a comprehensive analysis of the statistical attributes within the candidate solutions discovered during the player evaluation framework optimization process. Attributes are tracked and ranked based on their frequency of selection across candidate solutions, both globally and segmented by the number of clusters (K).

1. Attribute Frequency Ranking for K = 3

The following table lists the player performance features ranked by their frequency of selection within solutions optimized specifically under a 3-cluster configuration (K = 3).

Position	Attribute	Frequency
1st (tie)	G/SoT-Shooting	6 appearances
1st (tie)	Fld-Goal and Shot Creation	6 appearances
3rd	npG/Sh-Shooting	5 appearances
4th (tie)	Cmp%.1-Passing	4 appearances
4th (tie)	np:G-xG-Shooting	4 appearances
4th (tie)	90s-Defensive Actions	4 appearances

2. Attribute Frequency Ranking for K = 2

The following table details feature selection frequency for candidate solutions optimized under a 2-cluster configuration (K = 2).

Position	Attribute	Frequency
1st	90s-Defensive Actions	8 appearances
2nd (tie)	Fld-Goal and Shot Creation	5 appearances
2nd (tie)	Cmp%.1-Passing	5 appearances
4th (tie)	npG/Sh-Shooting	3 appearances
4th (tie)	np:G-xG-Shooting	3 appearances
6th	Cmp%-Passing	2 appearances
7th (tie)	Lost-Miscellaneous Stats	1 appearance

Position	Attribute	Frequency
7th (tie)	Dead-Pass Types	1 appearance
7th (tie)	TI-Pass Types	1 appearance

3. Global Attribute Frequency and Scope Levels

This section outlines the overall distribution of attribute selection across all candidate sets, alongside their tactical role **Scope Classification Level** (representing how well an attribute cuts across different position profiles sustainably).

Rank	Attribute	Total Freq.	K = 3	K = 2
1st	90s-Defensive Actions	12	4	8
2nd	Fld-Goal and Shot Creation	11	6	5
3rd	Cmp%.1-Passing	9	4	5
4th	np:G/Sh-Shooting	8	5	3
5th	np:G-xG-Shooting	7	4	3
6th	G/SoT-Shooting	6	6	0
7th	Cmp%-Passing	2	0	2
8th (tie)	Lost-Miscellaneous Stats	1	0	1
8th (tie)	Dead-Pass Types	1	0	1
8th (tie)	TI-Pass Types	1	0	1

Individual Attribute Scope Profile

- 90s-Defensive Actions: Level 4 Scope
- Fld-Goal and Shot Creation: Level 2 Scope
- Cmp%.1-Passing: Level 4 Scope
- np:G/Sh-Shooting: Level 2 Scope
- np:G-xG-Shooting: Level 2 Scope
- G/SoT-Shooting: Level 1 Scope
- Cmp%-Passing: Level 4 Scope
- Lost-Miscellaneous Stats: Level 4 Scope
- Dead-Pass Types: Level 3 Scope
- TI-Pass Types: Level 2 Scope

4. Candidate Solutions Ranked by Optimization Score

The global ranking of attribute trio combinations based on their clustering mathematical score. Solutions under $K = 2$ heavily dominate the absolute apex of optimization performance metrics.

Pos.	Solution Combination Trio	Score	Config
1st	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%.1-Passing']	5849.06	K=2
2nd	['Fld-Goal and Shot Creation', '90s-Defensive Actions', 'np:G/Sh-Shooting']	5794.70	K=2
3rd	['90s-Defensive Actions', 'np:G/Sh-Shooting', 'Cmp%.1-Passing']	5637.89	K=2
4th	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'np:G-xG-Shooting']	5575.86	K=2
5th	['90s-Defensive Actions', 'Cmp%.1-Passing', 'np:G-xG-Shooting']	5428.45	K=2
6th	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%-Passing']	5382.08	K=2
7th	['np:G/Sh-Shooting', '90s-Defensive Actions', 'np:G-xG-Shooting']	5376.29	K=2
8th	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Lost-Misc Stats']	5259.58	K=2
9th	['90s-Defensive Actions', 'Cmp%.1-Passing', 'Cmp%-Passing']	5249.43	K=2
10th	['G/SoT-Shooting', 'Fld-Goal and Shot Creation', 'np:G/Sh-Shooting']	5212.31	K=3
11th	['G/SoT-Shooting', 'np:G-xG-Shooting', 'np:G/Sh-Shooting']	5152.58	K=3
12th	['G/SoT-Shooting', 'Cmp%.1-Passing', 'np:G/Sh-Shooting']	5139.24	K=3
13th	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', '90s-Defensive Actions']	5118.47	K=3
14th	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'np:G/Sh-Shooting']	5045.34	K=3
15th	['Fld-Goal and Shot Creation', 'G/SoT-Shooting', 'np:G-xG-Shooting']	4990.36	K=3
16th	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', 'G/SoT-Shooting']	4984.39	K=3
17th	['Dead-Pass Types', 'Cmp%.1-Passing', 'TI-Pass Types']	4973.44	K=2

Pos.	Solution Combination Trio	Score	Config
18th	['Cmp%.1-Passing', 'np:G-xG-Shooting', 'G/SoT-Shooting']	4929.17	K=3
19th	['np:G/Sh-Shooting', '90s-Defensive Actions', 'Cmp%.1-Passing']	4856.37	K=3
20th	['90s-Defensive Actions', 'np:G-xG-Shooting', 'Fld-Goal and Shot Creation']	4771.80	K=3

5. Candidate Solutions Ranked by Total Scope

The global evaluation of solutions sorted by their composite **Total Scope**, illustrating how role-independent and flexible the metric configuration remains when tracking performance across highly diverse tactical positions.

Pos.	Solution Combination Trio	Total Scope	Score Rank	Config
1st	['90s-Defensive Actions', 'Cmp%.1-Passing', 'Cmp%-Passing']	12	9th	K=2
2nd (tie)	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%.1-Passing']	10	1st	K=2
2nd (tie)	['90s-Defensive Actions', 'npG/Sh-Shooting', 'Cmp%.1-Passing']	10	3rd	K=2
2nd (tie)	['90s-Defensive Actions', 'Cmp%.1-Passing', 'np:G-xG-Shooting']	10	5 ^o	K=2
2nd (tie)	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%-Passing']	10	6th	K=2
2nd (tie)	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Lost-Misc Stats']	10	8th	K=2
2nd (tie)	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', '90s-Defensive Actions']	10	13th	K=3
2nd (tie)	['npG/Sh-Shooting', '90s-Defensive Actions', 'Cmp%.1-Passing']	10	19th	K=3
9th	['Dead-Pass Types', 'Cmp%.1-Passing', 'TI-Pass Types']	9	17th	K=2
10th (tie)	['Fld-Goal and Shot Creation', '90s-Defensive Actions', 'npG/Sh-Shooting']	8	2nd	K=2
10th (tie)	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'np:G-xG-Shooting']	8	4th	K=2
10th (tie)	['npG/Sh-Shooting', '90s-Defensive Actions', 'np:G-xG-Shooting']	8	7th	K=2
10th (tie)	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'npG/Sh-Shooting']	8	14th	K=3
10th (tie)	['90s-Defensive Actions', 'np:G-xG-Shooting', 'Fld-Goal and Shot Creation']	8	20th	K=3
		7	12th	K=3

Pos.	Solution Combination Trio	Total Scope	Score Rank	Config
15th (tie)	['G/SoT-Shooting', 'Cmp%.1-Passing', 'np:G/Sh-Shooting']			
15th (tie)	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', 'G/SoT-Shooting']	7	16th	K=3
15th (tie)	['Cmp%.1-Passing', 'np:G-xG-Shooting', 'G/SoT-Shooting']	7	18th	K=3
18th (tie)	['G/SoT-Shooting', 'Fld-Goal and Shot Creation', 'np:G/Sh-Shooting']	5	10th	K=3
18th (tie)	['G/SoT-Shooting', 'np:G-xG-Shooting', 'np:G/Sh-Shooting']	5	11th	K=3
18th (tie)	['Fld-Goal and Shot Creation', 'G/SoT-Shooting', 'np:G-xG-Shooting']	5	15th	K=3

6. Analytical Visualization Mapping Overview

The corresponding repository directories contain the graphical analysis charts detailing the exact behavioral distributions modeled via K-Means:

- **Scatter Plot Cluster Layout (Highest Score Solution):** Diagnostic comparison visualizing the structural clusters mapped under $K = 2$ and $K = 3$ configurations for the absolute top-scoring combination.
- **Scatter Plot Cluster Layout (Highest Scope Solution):** Comparison layout validating the spatial density of clusters across $K = 2$ and $K = 3$ for the top-scope trio.